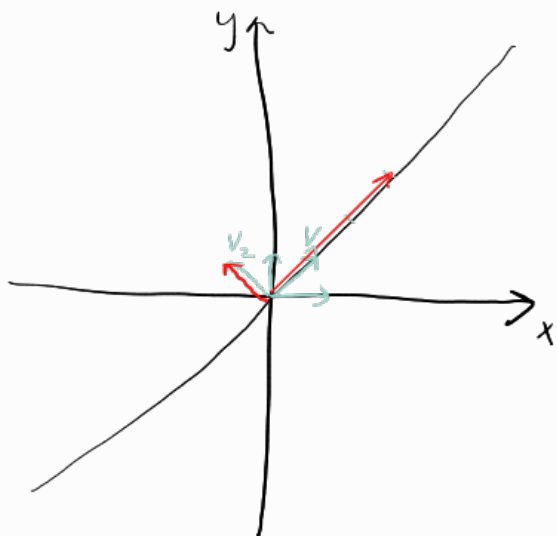


12

Zapiši matriko za razteg evklidske ravnine za faktor 3 v smeri premice $y=x$ z uporabo prehodne matrike



$$B = \{v_1, v_2\}$$

$$[v_1]_B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$D(v_1) = 3v_1$$

$$[v_2]_B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$D(v_2) = v_2$$

$$D_B^B = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$[v_1]_\varepsilon = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$[v_2]_\varepsilon = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$D_\varepsilon^\varepsilon = P_\varepsilon^B D_B^B P_B^\varepsilon =$$

$$P_\varepsilon^B = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} =$$

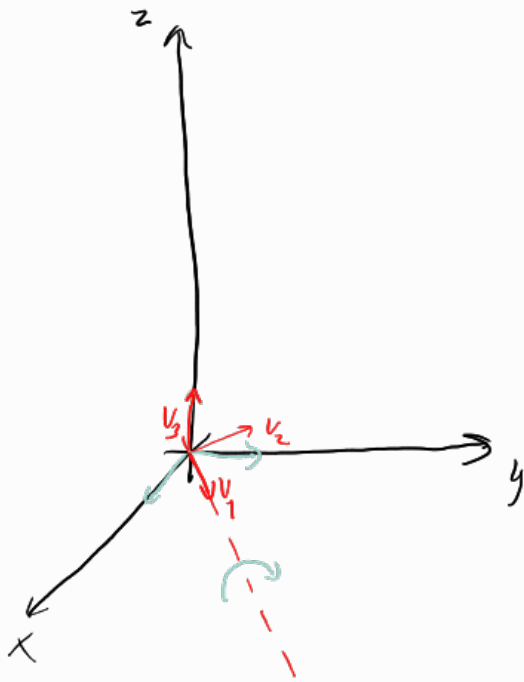
$$P_B^\varepsilon = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{3}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} =$$

$$= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

* simetrična matrika predstavlja razteg v $\mathbb{R}^n$ v paroma $\perp$ smernih.

13) Zapiši matriko za rotacijo za $\varphi = \frac{\pi}{2}$ okoli premice $\vec{r}(t) = t(1, 1, 0)$ z uporabo prehodnih matrik.



$$D(v_1) = v_1$$

$$D(v_2) = -v_3$$

$$D(v_3) = v_2$$

$$D_B^B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & \sin \varphi \\ 0 & -\sin \varphi & \cos \varphi \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$v_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

$$v_2 = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

$$v_3 = (0, 0, 1)$$

$$P_B^B = [v_1, v_2, v_3] = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P_B^E = (P_B^B)^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

ortogonalnost

$$A A^T = I$$

- enotski
- pravokotni

$$D_E^E = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} =$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

14) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ ve $T(1,1) = (0,1,2)$; $T(-1,1) = (2,1,0)$.

Doloci: $[T]_{\xi_3}^{\xi_2}$

$$\mathcal{B} = \{(1,1), (-1,1)\}$$

$$[T]_{\xi_3}^{\mathcal{B}} = \begin{bmatrix} 0 & 2 \\ 1 & 1 \\ 2 & 0 \end{bmatrix} \quad [P]_{\xi_2}^{\mathcal{B}} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \rightarrow [P]_{\mathcal{B}}^{\xi_2} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$[T]_{\xi_3}^{\xi_2} = \begin{bmatrix} 0 & 2 \\ 1 & 1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \underline{\underline{\begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}}}$$

Ker T linearna:

$$e_1 = \frac{v_1 - v_2}{2} \rightarrow T(e_1) = T\left(\frac{v_1 - v_2}{2}\right) = \frac{1}{2}(T(v_1) - T(v_2)) = \frac{1}{2}\left(\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$e_2 = \frac{v_1 + v_2}{2} \rightarrow T(e_2) = T\left(\frac{v_1 + v_2}{2}\right) = \frac{1}{2}(T(v_1) + T(v_2)) = \frac{1}{2}\left(\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\rightarrow \underline{\underline{[T]_{\xi_3}^{\xi_2} = \begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}}}$$

$$(15) \quad D: \mathbb{R}_2[x] \rightarrow \mathbb{R}_2[x]$$

$$D(p) = p'$$

$$T: \mathbb{R}_2[x] \rightarrow \mathbb{R}_2[x]$$

$$B = \{ \underset{v_1}{1}, \underset{v_2}{x+x^2}, \underset{v_3}{x-x^2} \}$$

$$[T]_B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\downarrow D$$

$$\{0, 1+2x, 1-2x\}$$

$$\text{Izračunaj } [DT]_B$$

$$\Sigma = \{1, x, x^2\}$$

$$[D]_{\Sigma} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[P]_{\Sigma}^B = \begin{bmatrix} [v_1]_{\Sigma} & [v_2]_{\Sigma} & [v_3]_{\Sigma} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$[P]_{\Sigma}^B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}^{-1} = \frac{1}{\det P_{\Sigma}^B} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$[D]_B^B = [P]_{\Sigma}^B [D]_{\Sigma}^B$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{bmatrix} =$$

$$= \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$[DT]_B^B = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & -1 \\ -1 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{im}(DT) = \text{Span}\{(v_1+v_2+v_3), (v_1-v_2-v_3)\} =$$

$$= \text{Span}\{(1+2x), (1-2x)\}$$

$$\text{Ker}(DT) = \text{Span}\{(-1+x-x^2)\}$$

$$\downarrow$$

$$a_1 + a_2 + a_3 = 0$$

$$a_2 = 0$$

$$a_1 = -a_3$$

$$p(x) = a_3(-1, 0, 1) = a_3(-v_1 + v_3) =$$

$$= a_3(-1+x-x^2)$$

